FTM IMCQ2
Fall 2025

Histology, Physiology, Biochemistry

KEY File

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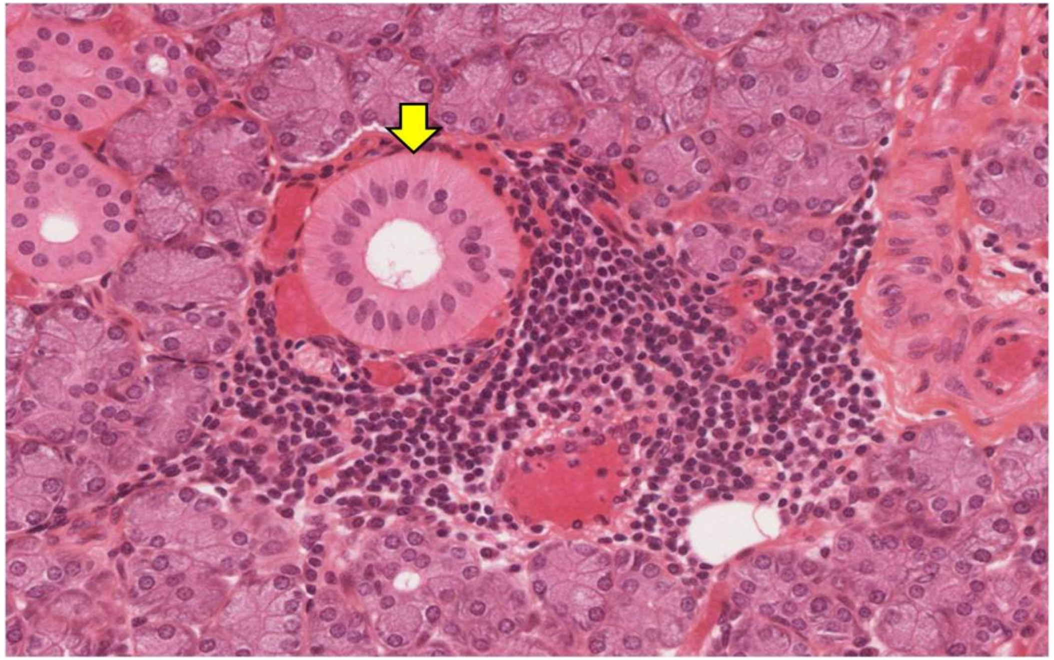
Histology Questions (4)

1. A 46-year-old man comes to the physician because of a 3-week history of increased viscosity of his saliva, and a swelling below the side of his jaw. Physical examination shows enlargement of the submandibular gland. A photomicrograph of a biopsy specimen obtained from this gland is shown; the arrow indicates a striated duct from the gland. This duct has many basal infoldings which aids it in its function. Which of the following is true in relation to the basal infoldings of this structure?

A. It is an interlobar duct that is closest to the oral cavity
B. It helps to increase the surface area for cell-cell exchange
C. It has abundant rER for protein secretion into the lumen
D. It is an apical specialization for maximum absorption from the lumen
E. It is an apical specialization involved in movement of particles on the cell's surface



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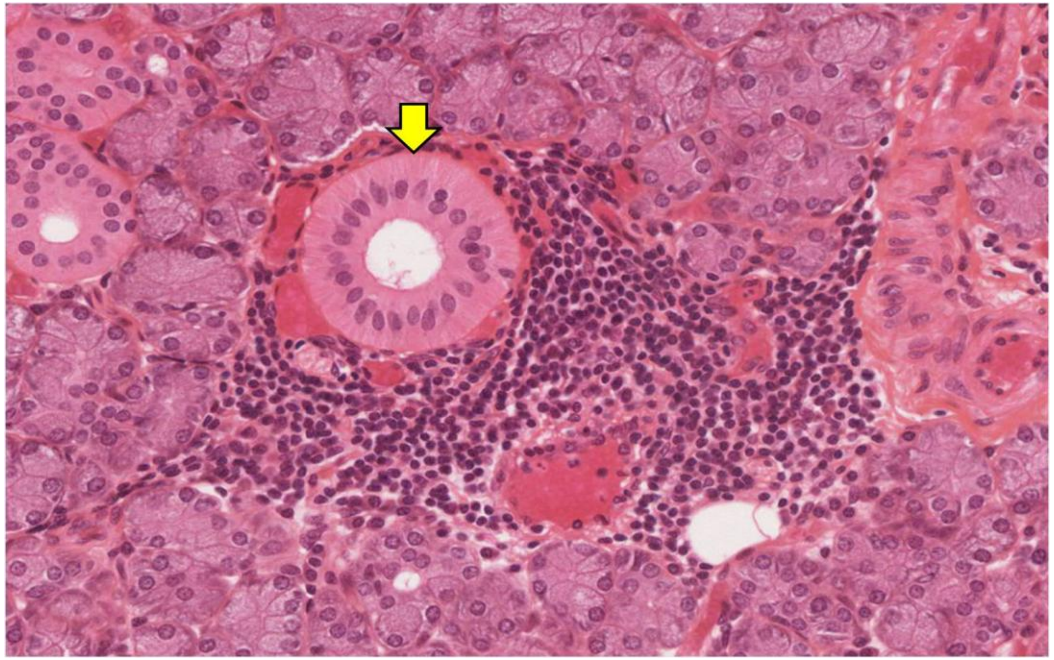

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Identify, define and describe the various types of exocrine glands (simple tubular glands, branched tubuloacinar serous and mucus glands, and serous demilunes). SOM.MK.I.BPM1.1.FTM.3.HCB.0814

2. A 49-year-old man was brought to the emergency department 2 hours after sustaining an open wound to his left finger. He appears to be in painful distress. His blood pressure is 122/80 mm Hg and pulse is 82/min. Physical examination shows a 1.5cm x 2cm lesion affecting the left index finger. Which of the following best characterizes the epithelial tissue most likely damaged in this patient?

A. Single layer of cells with rounded nuclei
B. Single layer of cells with nuclei that are taller than they are wide
C. Several layers of cells with a-nucleated cells at the top- most layer
D. Single layer of cells with nuclei at various levels
E. Multiple layer of cells with nuclei present at the top-most layer

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Differentiate between the different types of epithelia based on their structure, function and localization.
SOM.MK.I.BPM1.1.FTM.3.HCB.0774

3. A 25-year-old woman was brought to the physician because of a 3-day history of chest pain and joint aches, and a 1-day history of fevers, chills and shortness of breath. Her temperature is 40 ° C (104°F). Physical examination shows a heart murmur. Transesophageal echocardiogram confirms endocarditis (inflammation of the endocardial layer of the heart). This layer is often lined by an epithelium containing a single layer of cells with nuclei that are flattened, and attenuated cytoplasm. Which of the following best identifies this epithelium?

A. Simple squamous
B. Simple cuboidal
C. Simple columnar
D. Pseudostratified
E. Transitional
F. Stratified cuboidal

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4. A 58-year-old man is brought to the emergency department because of a 20-minute history of sustained seizure activity. He has a history of epilepsy. His blood glucose concentration is 55 mg/dL (normal range: 75mg/dL – 110 mg/dL). Physical examination shows a comatose man. He dies despite appropriate medical intervention. Examination of a microscopic specimen of his brain shows dysfunction of the junctional complex seen in the attached electron micrograph. Which of the following best characterizes the dysfunctional structure?

A. Composed of the protein connexin
B. Located on the basal aspect of the cell
C. Stabilized intracellularly by intermediate filaments
D. Consists of a protein called plakoglobin
E. Intracellularly linked to actin filaments

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Describe the histological structure of gap junctions.
SOM.MK.I.BPM1.1.FTM.3.HCB.0784

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Physiology Questions (3)

1. Nasal epithelial cells are isolated and bathed in a solution with electrolyte concentrations as shown in the table. The membrane potential is measured at -73 mV. Potassium is at normal physiological levels (120 mmol/L intracellular; 5 mmol/L extracellular), yielding a K^+ equilibrium potential of -83 mV. What is the most likely driving force and direction of flux for potassium?

Answer	Driving force	Direction
A	0 mV	no movement
B	+10 mV	out
C	-10 mV	in
D	+73 mV	out
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SOM.MK.II.BPM1.1.FTM.3.PHYS.0028 List the concentrations and equilibrium potentials of Na, K, Cl, Ca in a typical non-excitable cell. Based on the Nernst equilibrium potential, predict the direction of transmembrane flux for an ion when the membrane potential a) is at its equilibrium potential, b) is higher than the equilibrium potential, or c) is less than the equilibrium potential.

2. A myocardial cell is bathed in an electrolyte solution that yields the following equilibrium potentials:

$$E_{Na} = +80 \text{ mV}$$
$$E_{K} = -40 \text{ mV}$$
$$E_{Cl} = -40 \text{ mV}$$

During an action potential the membrane potential reaches a stable plateau of +20 mV. During this plateau phase, which ion conductance profile is most likely? (g = conductance; >> means much greater)

A. $g_{Cl} >> g_{K} = g_{Na}$
B. $g_{K} >> g_{Cl} = g_{Na}$
C. $g_{Na} >> g_{K} = g_{Cl}$
D. $g_{Cl} = g_{Na} >> g_{K}$
E. $g_{Na} = g_{Cl} = g_{K}$

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D. $g_{Cl} = g_{Na} >> g_K$
E. $g_{Na} = g_{Cl} = g_K$

SOM.MK.II.BPM1.1.FTM.3.PHYS.0030 Using the Goldman-Hodgkin-Katz equation, explain how the relative permeabilities of the 4 key ion species (Na^+, K^+, Cl^-, Ca^{2+}) create the resting membrane potential. Given an increase or decrease in the permeability for each of the ions, predict how the membrane potential would

3. A 9-year-old boy is brought to the emergency department because of 3-day history of progressive muscle cramps. Laboratory studies show the following serum electrolyte concentrations:

Na^+ = 139 mmol/L	(normal range 136 – 145 mmol/L)
K^+ = 3.7 mmol/L	(normal range 3.5 – 5 mmol/L)
Ca^{2+} = 1.6 mmol/L	(normal range 2.1 – 2.8 mmol/L)

Which alteration in voltage-gated ion channels is the most likely cause of the muscular cramps in this patient?

A. More Ca^{2+} channels open
B. Fewer Ca^{2+} channels open
C. More Na^+ channels open
D. Fewer Na^+ channels open
E. More K^+ channels open
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SOM.MK.II.BPM1.1.FTM.3.PHYS.0038 Draw a graph that shows how the voltage sensitivity of the voltage gated sodium channel changes with extracellular calcium concentration.

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Biochemistry Questions (9)

A researcher is studying bacterial DNA replication in *Escherichia coli*. She introduces a point mutation in the gene encoding DNA polymerase I that eliminates its 5′→3′ exonuclease activity while preserving its 3′→5′ exonuclease and polymerase functions. Which of the following steps in DNA replication will be most directly impaired in this mutant strain?

A. Proofreading newly synthesized DNA during elongation
B. Removal of RNA primers from Okazaki fragments
C. Synthesis of the RNA primers required for replication initiation
D. Unwinding of the parental DNA double helix
E. Prevention of DNA reannealing during replication

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0044 Describe the significance of the various proteins and enzymes required for replication and predict the effects of mutations of these proteins and enzymes on DNA replication.

A 40-year-old woman comes to the physician because of a 2-month history of abdominal pain, bloating and unexplained weight loss. A CT scan and an image-guided biopsy confirm a diagnosis of ovarian cancer. She is prescribed IV-treatment with a drug similar in action to camptothecin. Which of the following best describes the mechanism of action of the prescribed drug?

A. Inhibitor of DNA Polymerase ε, thus slowing cell proliferation
B. A nucleoside analogue that lacks 3'-OH group, thus blocking DNA strand elongation
C. A topoisomerase-II inhibitor, causing DNA double strand breaks and cytotoxicity
D. A telomerase inhibitor that ultimately causes senescence of tumour cells
E. A topoisomerase-I inhibitor, causing DNA single strand breaks
F. A helicase inhibitor, impairing strand separation

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0047; Outline the mechanisms of action of drugs that interfere with DNA replication
SOM.MK.I.BPM1.1.FTM.3.BCHM.0041; Describe the role of single-strand binding protein & topoisomerases in DNA replication

A 6-month-old boy is brought to the physician by his parents for a follow-up evaluation of poor feeding, hypotonia, and developmental delay. Laboratory studies show macrocytic anemia with elevated homocysteine levels. Molecular analysis shows a mutation in the consensus sequence containing GT bases at the start of intron 1 of the gene encoding cystathionine β-synthase (CBS). Which of the following best explains how this mutation leads to the patient’s clinical presentation?

A. Loss of the 7-methylguanosine cap prevents ribosome binding to the mRNA
B. Aberrant excision of introns results in retention of noncoding sequences in the mRNA
C. Failure to polyadenylate the mRNA causes premature degradation in the cytoplasm
D. Impaired transport of mature mRNA from the nucleus to the cytoplasm
E. Failure of the ribosome to recognize the correct AUG start codon during translation

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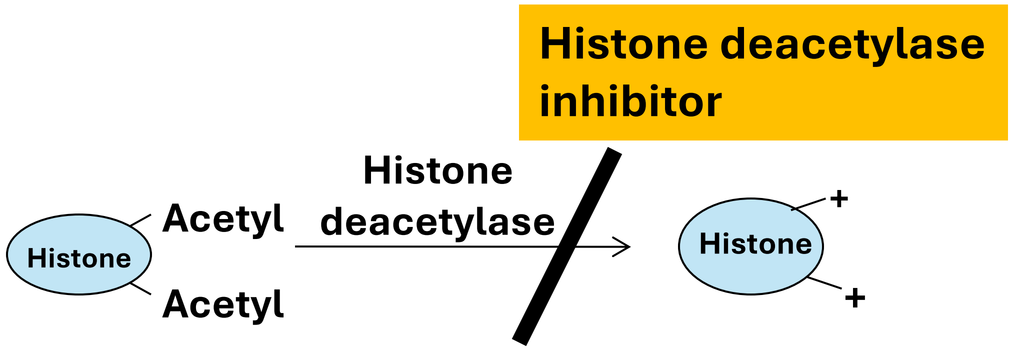
SOM.MK.I.BPM1.1.FTM.3.BCHM.0061 Predict effects of splice site mutations on mature mRNA and ultimately the translated protein

A pharmacist has discovered a compound 'X' that acts as a histone deacetylase inhibitor. Which of the following best describes the predicted effect of this drug (when compared to absence of drug)?



A. Decreases acetylation of histone
B. Increases binding of histone to DNA
C. Increases positive charge on histone
D. Favors formation of heterochromatin
E. Increases expression of the genes

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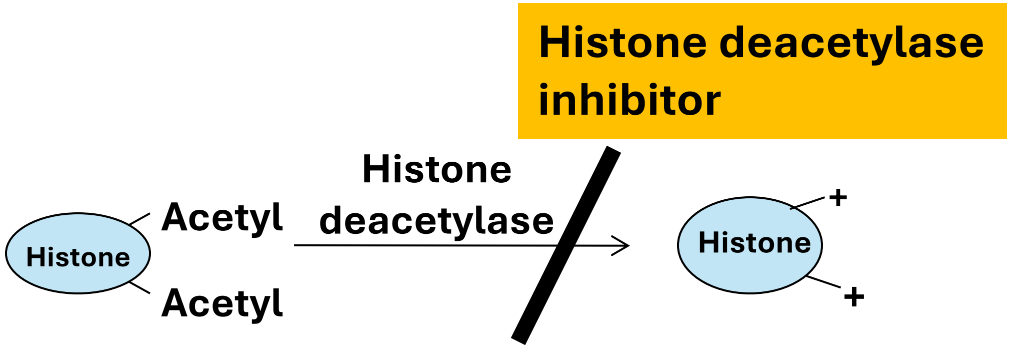

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0034; Describe how histone modification can influence transcriptional activity

A researcher is examining chromatin structure in eukaryotic cells using electron microscopy. In one experiment, she selectively removes a protein that binds to the DNA between nucleosomes. After removal, the chromatin adopts a more “beads-on-a-string” appearance, with decreased compaction into higher-order fibers. Which of the following proteins was most likely removed in this experiment?

A. Histone H2A
B. Histone H2B
C. Histone H3
D. Histone H4
E. Histone H1

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0033 Describe the different structural conformations of DNA packaging in eukaryotes (10 nm fiber, 30 nm fiber, and metaphase chromosomes)

A 54-year-old man is undergoing chemotherapy for acute myeloid leukemia. His regimen includes cytarabine (ara-C), which must first be activated by intracellular kinases. Once activated, the drug incorporates into DNA during replication and inhibits DNA polymerase, leading to chain termination. Which of the following best describes the chemical classification of this drug?

A. Nucleotide analog that is demethylated upon entry into the cell
B. Nucleotide analog that requires removal of its phosphate group to be active
C. Nucleoside analog that requires phosphorylation to be active
D. Nucleoside analog that requires acetylation to be active
E. Nucleotide analog possess three phosphate groups

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0047 Outline the mechanisms of action of drugs that interfere with DNA replication

A 9-year-old girl is brought to the physician by her parents for a follow-up examination because of a 6-month history of chronic infections, eczema, and thrombocytopenia. Laboratory studies show reduced expression of several cytokine genes. A molecular biologist performs a nuclear run-on transcription assay using isolated nuclei from the patient's lymphocytes. The results show that RNA polymerase II is recruited to promoters but fails to elongate mRNA beyond the first few nucleotides. Sequencing of the patient's DNA identifies a missense mutation in a gene encoding a component of the general transcription factor TFIIH. Which of the following steps in eukaryotic transcription is most directly affected by this mutation?

A. Formation of the preinitiation complex and promoter recognition
B. Phosphorylation of RNA polymerase II
C. Capping of the nascent mRNA at the 5' end
D. Splicing of pre-mRNA introns
E. Termination of transcription at polyadenylation signals

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0044 Describe the significance of the various proteins and enzymes required for replication and predict the effects of mutations of these proteins and enzymes on DNA replication

A 2-year-old girl is brought to the physician by her parents because of a 3-month history of pale appearance and inability to play with her siblings as she tires easily. Laboratory studies show very low hemoglobin levels, elevated HbF levels and no detectable HbA. Additional studies show significantly reduced levels of β-globin transcripts. The β-globin transcripts are less than half the length of the normal β-globin mRNA. Which of the following best explains the molecular reason for these observations?

A. Impaired capping of β-globin mRNA
B. Covalent modification of the β-globin protein
C. Improper folding of the β-globin protein
D. Promoter mutations in the β-globin gene
E. Splicing alteration of the β-globin mRNA

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0061; Predict effects of splice site mutations on mature mRNA and ultimately the translated protein

A 10-year-old boy is brought to the physician by his mother because of a 3-month history of poor adjustment to school. Two maternal uncles are also similarly affected. Neurological examination shows mild intellectual disability. Further studies show a point mutation in CSTF gene resulting in a loss of function of the cleavage stimulating factor-F (CstF). Which of the following best describes the mechanistic consequence of this mutation in the patient’s cells?

A. Decreased efficiency of 5’-capping of mRNA
B. Increased rate of transcriptional initiation
C. Decreased efficiency of 3-polyadenylation of mRNA
D. Increased translation of mRNA
E. Increased nuclear export of mRNA
F. Impaired post-transcriptional modifications in t-RNA

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0060; Describe the post-transcriptional modifications of eukaryote mRNA and the significance of these modifications (5' capping, polyadenylation, and splicing)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0063; Describe human disease conditions that may be associated with mRNA modification and errors in this process